

Processing Dangerous Paths – On Security and Privacy of the Portable Document Format

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PDF is the de-facto standard for document exchange. It is common to open PDF files from potentially untrusted sources such as email attachments or downloaded from the Internet. In this work, we perform an in-depth analysis of the capabilities of malicious PDF documents. Instead of focusing on implementation bugs, we abuse legitimate features of the PDF standard itself by systematically identifying dangerous paths in the PDF file structure. These dangerous paths lead to attacks that we categorize into four generic classes: (1) Denial-of-Service attacks affecting the host that processes the document. (2) Information disclosure attacks leaking personal data out of the victim's computer. (3) Data manipulation on the victim's system. (4) Code execution on the victim's machine. An evaluation of 28 popular PDF processing applications shows that 26 of them are vulnerable at least one attack. Finally, we propose a methodology to protect against attacks based on PDF features systematically.

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